Assignment- linear regression

Question 1: What is Simple Linear Regression?

Ans- Simple Linear Regression is a statistical method used to model the relationship between two variables:

- One independent variable (X)
- One dependent variable (Y)

It assumes that the relationship between X and Y can be represented by a straight line.

$$Y=a+bX$$

Question 2: What are the key assumptions of Simple Linear Regression?
Ans:
Simple Linear Regression is based on a few important assumptions. These assumptions help ensure that the model gives accurate and reliable results:-
1. there should be a **linear relationship** between the independent variable and the dependent variable. This means that when the value of X changes, the value of Y should change in a straight-line pattern.
2. the observations must be **independent of each other**. The value of one observation should not influence or depend on another.
3. the model assumes **homoscedasticity**, which means the errors or residuals should have a constant variance across all values of the independent variable.

Another assumption is that the **residuals are normally distributed**. This helps in making correct predictions and statistical inferences.

Lastly, since it is a simple linear regression model, there should be **only one independent variable**, so the issue of multicollinearity does not arise.

Question 3: What is heteroscedasticity, and why is it important to address in regression models?

Ans:- Heteroscedasticity refers to a situation in a regression model where the variance of the error terms is not constant across all values of the independent variable. it means that the spread of residuals changes as the value of X increases or decreases, instead of remaining the same.

When heteroscedasticity is present, the regression model still gives coefficient estimates, but these estimates become less reliable. The standard errors may be incorrect, which can lead to wrong conclusions during hypothesis testing, such as misleading t-tests and confidence intervals.

It is important to address heteroscedasticity because it affects the **accuracy and validity** of the regression results. If it is ignored, the model may appear to be significant when it is not, or vice versa. To handle this problem, techniques such as data transformation, weighted least squares, or using robust standard errors can be applied.

Conclusion, detecting and correcting heteroscedasticity helps improve the **trustworthiness of regression analysis**.

Question 4: What is Multiple Linear Regression?

Ans:- Multiple Linear Regression is an extension of simple linear regression. While simple linear regression deals with one independent variable, multiple linear regression is used when there are two or more independent variables that may affect a dependent variable. ndent variable influences it.

$$Y= a+b_1X_1+b_2X_2+...+b_nX_n$$

Goal:-to **predict the dependent variable** and understand how each independent.

Question 5: What is polynomial regression, and how does it differ from linear regression?

Ans:- Polynomial Regression is a type of regression analysis where the relationship between the independent variable(s) and the dependent variable is modeled as a **polynomial (curved) relationship** rather than a straight line. It is useful when data shows a **non-linear pattern** that a simple linear regression cannot capture.

Polynomial regression is different from linear regression because it can model **curved relationships** between variables, while linear regression can only model a **straight-line relationship**.

In linear regression, the change in the dependent variable is assumed to be **proportional to the independent variable**, meaning the effect is constant. For example, if studying 1 extra hour always increases marks by 2 points, that's linear.

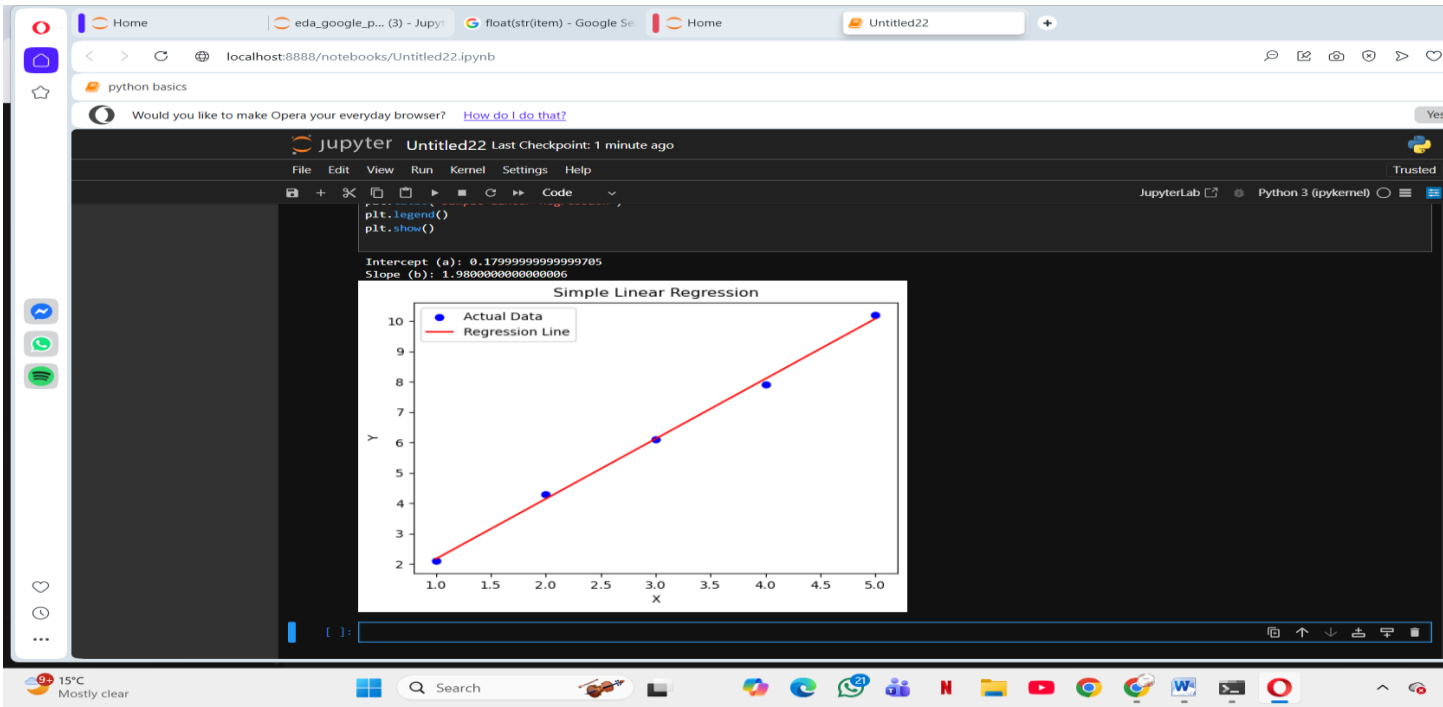
In polynomial regression, the effect of the independent variable on the dependent variable can **change at different levels**. For example, a plant's growth may speed up, then slow down, and then plateau over time—something a straight line cannot capture.

Question 6: Implement a Python program to fit a Simple Linear Regression model to the following sample data:

- X = [1, 2, 3, 4, 5]
- Y = [2.1, 4.3, 6.1, 7.9, 10.2]

Plot the regression line over the data points.
(Include your Python code and output in the code box below.)

ANS:-
```
# Import required libraries
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
import numpy as np

# Sample data
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)  # Independent variable (reshape for sklearn)
Y = np.array([2.1, 4.3, 6.1, 7.9, 10.2])     # Dependent variable

# Create the Linear Regression model
model = LinearRegression()

# Fit the model to the data
model.fit(X, Y)

# Make predictions using the fitted model
Y_pred = model.predict(X)

# Print the model coefficients
print(f"Intercept (a): {model.intercept_}")
print(f"Slope (b): {model.coef_[0]}")

# Plot the data points
plt.scatter(X, Y, color='blue', label='Actual Data')

# Plot the regression line
plt.plot(X, Y_pred, color='red', label='Regression Line')

# Labels and title
plt.xlabel('X')
plt.ylabel('Y')
plt.title('Simple Linear Regression')
plt.legend()
plt.show()
```



Question 7: Fit a Multiple Linear Regression model on this sample data:

- Area = [1200, 1500, 1800, 2000]
- Rooms = [2, 3, 3, 4]
- Price = [250000, 300000, 320000, 370000]

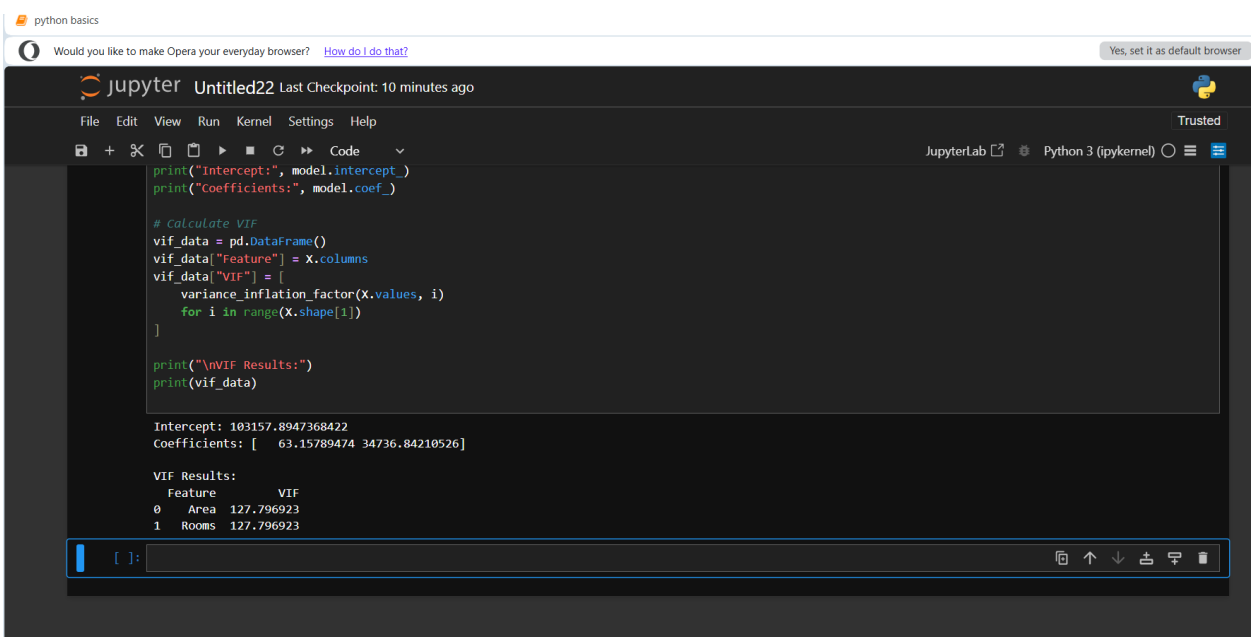
Check for multicollinearity using VIF and report the results.

(Include your Python code and output in the code box below.)

ANS:-

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from statsmodels.stats.outliers_influence import variance_inflation_factor

# Create the dataset
data = {
    "Area": [1200, 1500, 1800, 2000],
    "Rooms": [2, 3, 3, 4],
    "Price": [250000, 300000, 320000, 370000]
}
```

```
df = pd.DataFrame(data)

# Independent variables
X = df[["Area", "Rooms"]]

# Dependent variable
Y = df["Price"]

# Fit Multiple Linear Regression model
model = LinearRegression()
model.fit(X, Y)

# Print coefficients
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)

# Calculate VIF
vif_data = pd.DataFrame()
vif_data["Feature"] = X.columns
vif_data["VIF"] = [
    variance_inflation_factor(X.values, i)
    for i in range(X.shape[1])
]

print("\nVIF Results:")
print(vif_data)
```



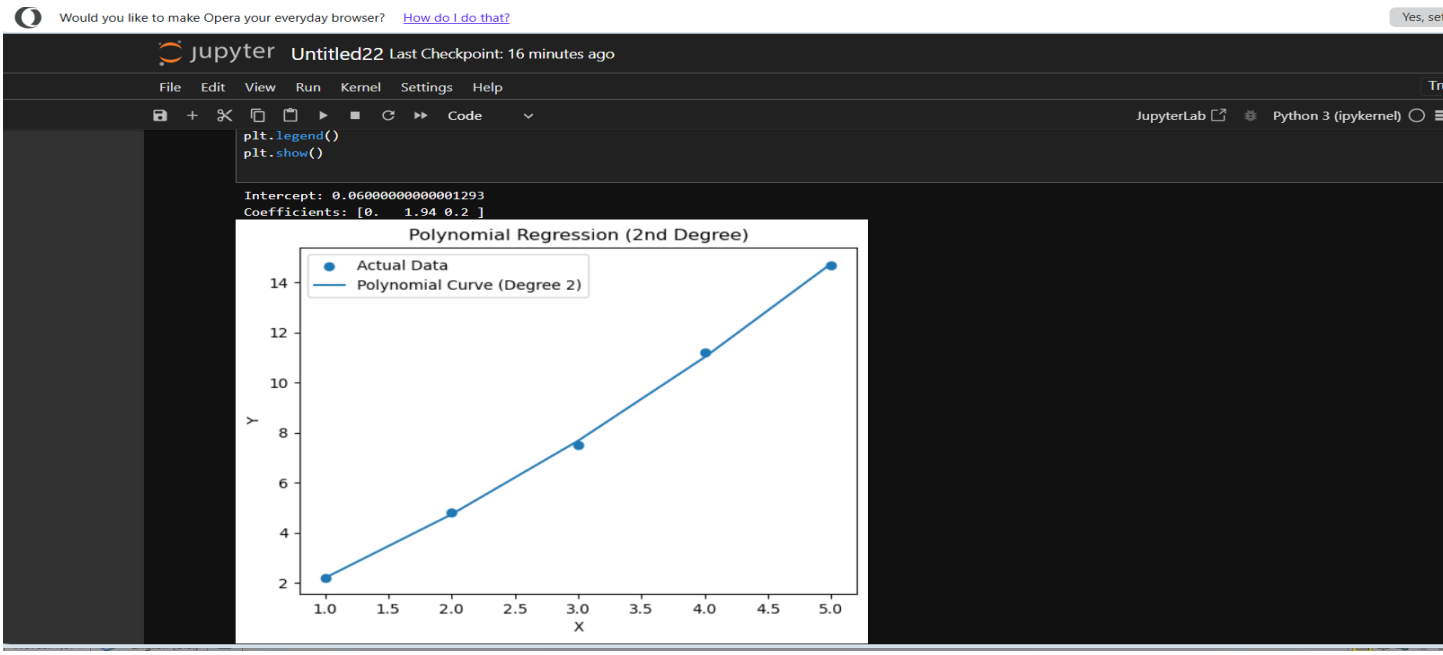
Question 8: Implement polynomial regression on the following data:

- X = [1, 2, 3, 4, 5]
- Y = [2.2, 4.8, 7.5, 11.2, 14.7]

Fit a 2nd-degree polynomial and plot the resulting curve.
(Include your Python code and output in the code box below.)

ANS:-

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression

# Given data
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
Y = np.array([2.2, 4.8, 7.5, 11.2, 14.7])

# Convert X into polynomial features (degree = 2)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

# Fit Polynomial Regression model
model = LinearRegression()
model.fit(X_poly, Y)

# Predict values
Y_pred = model.predict(X_poly)

# Print model coefficients
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)

# Plot original data points
plt.scatter(X, Y, label="Actual Data")

# Plot polynomial curve
plt.plot(X, Y_pred, label="Polynomial Curve (Degree 2)")

# Labels and title
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Polynomial Regression (2nd Degree)")
plt.legend()
plt.show()
```

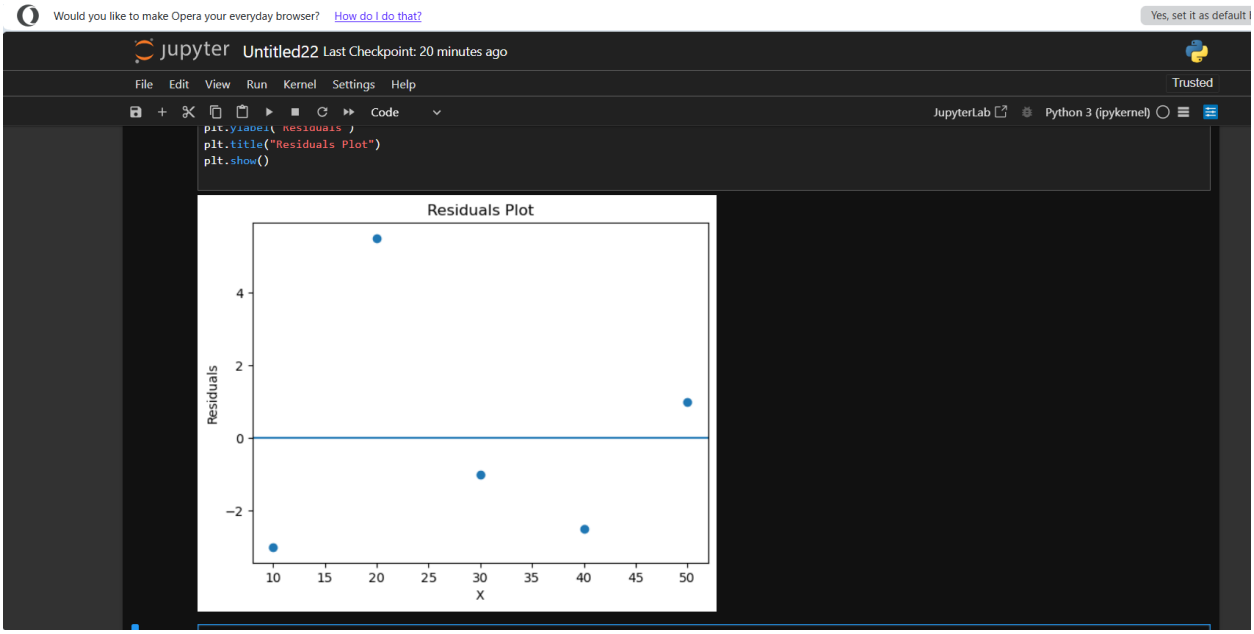


Question 9: Create a residuals plot for a regression model trained on this data:

- X = [10, 20, 30, 40, 50]
- Y = [15, 35, 40, 50, 65]

Assess heteroscedasticity by examining the spread of residuals. (Include your Python code and output in the code box below.

ANS:-

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Given data
X = np.array([10, 20, 30, 40, 50]).reshape(-1, 1)
Y = np.array([15, 35, 40, 50, 65])

# Train Simple Linear Regression model
model = LinearRegression()
model.fit(X, Y)

# Predicted values
Y_pred = model.predict(X)

# Calculate residuals
residuals = Y - Y_pred

# Create residuals plot
plt.scatter(X, residuals)
```

```
plt.axhline(y=0)  # Reference line at zero
plt.xlabel("X")
plt.ylabel("Residuals")
plt.title("Residuals Plot")
plt.show()
```